Rahul Juluru

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EXPERIENCE

University of Oklahoma **Aug 2024 - Present**
Software Engineer (GRA) *Norman, OK*
Distributed Systems, Cloud Infrastructure, Full-Stack Engineering

- Containerized a 5-stage manual data preprocessing pipeline into GCP-deployed microservices using Docker and Kubernetes, eliminating 70% of manual effort and cutting average ingestion time from ~ 4 hrs to under 40 min for a 3-researcher team.
- Slashed asynchronous task failure rate by 40% by instrumenting retry-based orchestration, schema validation, and structured logging across the distributed ingestion pipeline - reducing silent data drops from ~ 12% to under 2%.
- Decomposed monolithic preprocessing scripts into 4 independently versioned modules (ingest, transform, feature compute, export) using object-oriented design principles, enabling isolated per-stage iteration and cutting integration regression time by ~ 60%.

Neospark **Jun 2023 - Jun 2024**
Software Engineering *Hyderabad, India*

- Shipped a self-serve internal reporting portal (Node.js / Express / React) using RESTful API design that replaced ad-hoc spreadsheet workflows across 5+ teams - reclaiming ~ 8 hrs/week of analyst time and eliminating a 2-day manual reporting cycle.
- Accelerated multi-report dashboard load time by 25% (1.8s → 1.35s, within the 1.5s SLA target) by batching asynchronous API calls and eliminating redundant sequential fetches in the Express response layer.
- Refactored frontend state management from prop-drilling to Redux and Context API, reducing wasted renders from ~ 340ms to ~ 95ms per interaction (profiled via React DevTools) and raising Lighthouse performance score from 61 to 84.

Salesforce **Jan 2023 - May 2023**
Software Engineering *Remote*

- Engineered 12 Apex triggers and Flow automations to unify cross-object business logic across 3 Salesforce orgs - eliminating a class of manual data-reconciliation errors that caused ~ 30% of support escalations.
- Composed 4 Lightning Web Components with asynchronous RESTful bindings and optimized SOQL queries, cutting data retrieval latency by 40% and eliminating full-page refresh cycles for the sales dashboard.
- Hardened least-privilege access across 6 custom objects using Salesforce profiles and permission sets - reducing unauthorized field-access incidents to zero and passing a compliance security audit covering 200+ user records.

TECHNICAL SKILLS

- **Languages:** Java, Python, C/C++, SQL, JavaScript, TypeScript, HTML/CSS
- **Frameworks & Libraries:** React.js, Next.js, Node.js, Express.js, Spring Boot, FastAPI, Django, Angular
- **Infrastructure & DevOps:** Docker, Kubernetes, AWS, GCP, Redis, PostgreSQL, MongoDB, GitHub Actions, Jenkins, CI/CD Pipelines
- **Concepts:** Distributed Systems, Microservices Architecture, REST APIs, GraphQL, Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design, Load Testing, Observability & Monitoring, Agile/Scrum

SOFTWARE PROJECTS

Smart-Expense-Tracker - AI-Powered Expense Management System **Jan 2025 - Apr 2025**

- Architected a microservices-based system with 3 independently deployable services (Next.js frontend, Express.js API gateway, FastAPI AI service) to decouple ML model updates from UI releases - enabling zero-downtime service-level deployments without shared-state coupling.
- Launched an OCR-to-LLM receipt processing pipeline (Tesseract, OpenCV, GPT-4 categorization) achieving 92% classification accuracy across 15 expense categories - eliminating 60% of manual data entry validated over 200+ receipts.
- Scaled backend throughput from 120 to 240 RPS under sustained load by introducing Redis response caching and PostgreSQL query optimization with composite indexing - validated via k6 load tests simulating 500 concurrent users.

CogniFetch - NLP-Driven Syllabus-to-Resource Mapper **Aug 2024 - Oct 2024**

- Designed an OCR-to-NLP extraction pipeline (Tesseract, OpenNLP) that structured 10–20+ topics per syllabus upload with under 5% topic-boundary error across 5+ document formats - transforming unstructured PDFs into ranked, indexed query sets.
- Resolved a query-ambiguity problem in syllabus parsing by combining LLM-generated search queries with Google Search API relevance ranking - cutting average topic lookup time by ~ 50% across 30 diverse syllabus documents in end-to-end testing.
- Deployed a modular Flask backend with asynchronous task queuing, RESTful endpoints, and concurrent multi-format upload support - maintaining sub-2-second p95 response time under load testing with 20 simultaneous users.

EDUCATION

University of Oklahoma **Aug 2024 - May 2026**
Master of Science, Computer Science (GPA: 3.78/4.0) *Norman, OK*

CVR College of Engineering **Aug 2020 - May 2024**
Bachelor of Technology, Information Technology (GPA: 9.35/10.0 (Distinction, top 1%)) *Hyderabad, India*